FedOrbit: Adaptive Personalized Federated Learning for Non-IID LEO Satellite Constellations

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***Abstract*—Federated learning (FL) in Low Earth Orbit (LEO) satellite constellations is affected by non-IID data and irregular ground-station visibility, both driven by orbital geometry. Global aggregation performs poorly when orbit-level class distributions are disjoint, while strong personalisation can be excessive when these distributions overlap. We present FedOrbit, which combines continuous orbit-level training over inter-satellite links, class-aware hierarchical aggregation, quality-weighted feature aggregation with return-rate dampening, and adaptive feature decomposition based on inter-orbit class similarity. Across three remote-sensing benchmarks and two non-IID partitions, FedOrbit achieves the highest accuracy in five of six settings and is within 0.9 percentage points of the best result in the sixth. The gains over the strongest baseline reach 16.1 percentage points under Dirichlet partitioning and 8.6 under pathological partitioning, with the smallest per-orbit accuracy spread in five of six settings.**

***Index Terms*—Federated learning, LEO satellite constellations, personalised aggregation, non-IID data, non-terrestrial networks.**

I. Introduction

Low Earth Orbit (LEO) satellite constellations are now an important platform for global Earth observation [1], [2], and on-board processing makes it possible to train and run machine learning models directly on the satellite [3], [4]. Transmitting raw observation data to a ground station for centralised training is impractical given the intermittent nature of satellite-to-ground links, data privacy requirements, and bandwidth constraints. Federated learning (FL) [5] reduces the need for raw-data transfer: each satellite trains locally on its own observations and exchanges only model parameters with a ground station (GS).

FL in LEO constellations differs fundamentally from the terrestrial setting. In a Walker Delta constellation, each orbit follows a distinct ground track and observes a different geographic region, producing orbit-level class distributions that are inherently non-IID [6], [7]. Orbit-to-GS visibility is governed by the same orbital geometry, producing imbalanced participation. In the constellation considered in this work, approximately 47% of training rounds have no orbit visible to the GS, and the least-visited orbit participates in fewer than one-third as many rounds as the most-visited orbit [8]. As the class distribution of an orbit is linked to its ground track and its visibility schedule is determined by orbital geometry, class contribution and timing are systematically correlated rather than independent. This correlation between data heterogeneity and participation is the defining property of the LEO FL setting.

This paper presents FedOrbit, a personalised FL framework that addresses the correlation between data and participation in LEO constellations. The contributions are: (i) identification of two failure modes induced by this correlation, namely global-model collapse on pathological partitioning and over-personalisation on Dirichlet partitioning; (ii) *continuous orbit-level training* over inter-satellite links, so every orbit refines the model in every round; (iii) *class-aware hierarchical aggregation*, routing classifier updates by per-class data ownership; (iv) *quality-weighted feature aggregation with return-rate dampening*, limiting the influence of an orbit returning from a long absence; (v) *adaptive feature decomposition*, blending each orbit's personal feature extractor toward the global one at a rate set by inter-orbit class similarity. We evaluate FedOrbit on three remote-sensing classification benchmarks under both pathological and Dirichlet ($\alpha = 0.5$) partitions, with the largest gains on the partition where existing methods perform worst.

II. Related Work

FL under non-IID data. FedAvg [5] degrades under heterogeneous distributions [9]; FedProx [10], SCAFFOLD [11], and FedNova [12] use proximal regularisation, control variates, or local-step normalisation. These methods produce a single global model but do not model the coupling between local data and visibility-dependent participation.

Personalised FL. Ditto [13] adds an ℓ_2 penalty toward the global model; APFL [14] interpolates local and

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global models; other methods use regularised inner loops [15], meta-learned initialisations [16], or split representations [17]. These methods do not account for bias in the global reference model caused by visibility-dependent participation.

FL in non-terrestrial networks. Prior LEO FL work has addressed scheduling under irregular visibility [8], semi-asynchronous aggregation [18], aerial parameter servers [19], [20], decentralised intra-orbit aggregation [21], over-the-air computation [22], system heterogeneity [23], and decentralised personalisation [24]. The common focus is communication architecture, convergence speed, and scheduling. To our knowledge, prior work does not jointly address orbit-level non-IID data and visibility-dependent participation or adapt personalisation to inter-orbit similarity.

III. System Model

A. LEO Constellation and Visibility Model

We consider a Walker Delta constellation with parameters following prior FL-over-LEO work [8]: $L = 5$ orbits, $N_l = 4$ satellites per orbit (20 satellites total), altitude $h = 550\,\mathrm{km}$, inclination $i = 53°$, GS at latitude $\phi_g = 51°\,\mathrm{N}$, minimum elevation $\alpha_e = 10°$, and round duration $\Delta t = 5\,\mathrm{min}$. The GS acts as the parameter server.

Each satellite follows a circular Keplerian orbit with mean motion $n_o = \sqrt{\mu/(R_E + h)^3}$, where $\mu = 3.986 \times 10^5\,\mathrm{km}^3/\mathrm{s}^2$ is Earth's gravitational parameter and $R_E = 6371\,\mathrm{km}$ is Earth's mean radius. The right ascension of the ascending node (RAAN) of orbit l is $\Omega_l = 2\pi l/L$ for $l = 0, \ldots, L-1$, giving equally spaced orbits. The GS position in the Earth-centred inertial (ECI) frame rotates with Earth at angular velocity $\omega_E = 7.292 \times 10^{-5}$ rad/s. Satellite k in orbit l is visible from the GS when

$$\varepsilon_k(t) = \arcsin\left(\frac{(\mathbf{r}_k(t) - \mathbf{r}_g(t)) \cdot \hat{\mathbf{r}}_g(t)}{\|\mathbf{r}_k(t) - \mathbf{r}_g(t)\|}\right) \geq \alpha_e\,, \quad (1)$$

where $\mathbf{r}_k(t)$ and $\mathbf{r}_g(t)$ are the satellite and GS positions in the ECI frame and $\hat{\mathbf{r}}_g(t)$ is the local zenith unit vector at the GS. An orbit is treated as visible at round t if at least one of its satellites satisfies (1), since the orbit-level model is what is uploaded to the GS. A binary matrix $\mathbf{V} \in \{0,1\}^{T \times L}$ records visibility, with $V_{t,l} = 1$ if orbit l is visible at round t and 0 otherwise; the set of visible orbits at round t is $\mathcal{V}(t) = \{l : V_{t,l} = 1\}$. All evaluated methods use the same $\mathbf{V}$.

For $T = 400$, approximately 47% of rounds have no visible orbit, the per-orbit visibility shares are $15.2\%, 10.2\%, 8.5\%, 5.2\%$, and 14.0%, and the longest contiguous interval with no visible orbit is $\tau^{\max} = 191$ rounds.

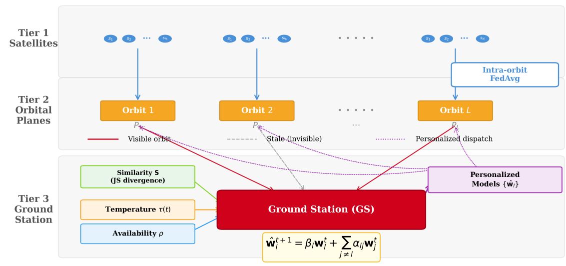

Fig. 1. FedOrbit architecture. Satellites train locally, aggregate within each orbit over ISLs, and downlink the orbit-level model to the GS during visibility windows.

B. Hierarchical Federated Learning Architecture

The architecture has satellite, orbit, and GS tiers (Fig. 1) and is shared by all methods. The following FedAvg-based procedure is used by the baselines. FedOrbit modifies both aggregation tiers and trains non-visible orbits over inter-satellite links, as described in Section V.

Local training. Satellite k holds a dataset $\mathcal{D}_k$ of size n_k with local objective $F_k(\mathbf{w}) = (1/n_k) \sum_{x \in \mathcal{D}_k} f(x; \mathbf{w})$, and runs E epochs of mini-batch SGD with learning rate $\eta_t = \eta_0 \gamma^t$, producing

$$\mathbf{w}_k^t = \texttt{LocalSGD}\big(\mathbf{w}^t, \mathcal{D}_k, E, \eta_t\big). \quad (2)$$

Intra-orbit aggregation. Satellites in orbit l combine their updates over short-range ISLs by sample-weighted averaging:

$$\mathbf{w}_l^t = \sum_{k \in \mathcal{S}_l} \frac{n_k}{\sum_{j \in \mathcal{S}_l} n_j} \mathbf{w}_k^t, \quad (3)$$

where $\mathcal{S}_l$ is the set of satellites in orbit l.

Inter-orbit aggregation at the GS. Baselines apply FedAvg [5] across the visible orbits:

$$\mathbf{w}^{t+1} = \sum_{l \in \mathcal{V}(t)} \frac{n_l}{\sum_{j \in \mathcal{V}(t)} n_j} \mathbf{w}_l^t, \quad (4)$$

with $n_l = \sum_{k \in \mathcal{S}_l} n_k$. FedProx, APFL, and Ditto use the same weighting and differ from FedAvg only in their local objectives.

C. Non-IID Data Distribution Model

Each orbit collects imagery along its ground track, producing a class distribution P_l that reflects the terrain it overflies. Because satellites in the same orbit share the same ground track, heterogeneity is orbit-specific rather than satellite-specific. Two partitioning methods are considered to represent contrasting levels of inter-orbit overlap:

Pathological partitioning. Each orbit is assigned a disjoint subset of classes, modelling orbits that observe disjoint land-cover categories. Within each orbit, satellite allocations follow a Dirichlet($\alpha = 0.5$) split.

Dirichlet partitioning ($\alpha = 0.5$). Inter-orbit class proportions are drawn from Dirichlet(0.5), giving overlapping but skewed orbit distributions. Intra-orbit allocation also follows Dirichlet(0.5). This models the partial overlap that arises when orbits cover geographically proximate but distinct regions.

IV. Problem Formulation

In the Walker Delta constellation of Section III, the class distribution P_l is linked to the ground track of orbit l, while the visibility schedule $\{V_{t,l}\}_{t=1}^{T}$ is determined by orbital geometry through (1). The standard FL assumption that client participation is independent of local data [5], [10], [11] therefore does not hold, and the global model cannot serve as an unbiased reference for personalisation.

Let $\{\widetilde{\mathbf{w}}_l\}_{l=1}^{L}$ denote the orbit-level inference models, F_l the expected loss on P_l, and Acc_l the corresponding accuracy. FedOrbit is guided by the following fairness-constrained objective:

$$\min_{\{\widetilde{\mathbf{w}}_l\}} \frac{1}{L}\sum_{l=1}^{L} F_l(\widetilde{\mathbf{w}}_l) \quad \text{s.t.} \quad \max_l \mathrm{Acc}_l - \min_l \mathrm{Acc}_l \le \delta. \tag{5}$$

Equation (5) expresses the design objective; the fairness term is not explicitly enforced during optimisation. The empirical spread is reported in Section VII.

The formulation imposes four design requirements: (**R1**) reduce contribution imbalance caused by $\mathbf{V}$; (**R2**) weight classifier updates by class ownership; (**R3**) adapt personalisation to inter-orbit heterogeneity without manual retuning; and (**R4**) keep GS-uplink traffic comparable to FedAvg, with additional work performed on-board or within an orbit.

V. Proposed Method: FedOrbit

FedOrbit modifies on-board scheduling, intra-orbit and inter-orbit aggregation, and per-orbit inference. The model is split into a feature extractor and a classifier, $\mathbf{w} = [\mathbf{w}^{\mathrm{f}}, \mathbf{w}^{\mathrm{c}}]$, since the feature extractor encodes content shared across the constellation while each classifier row is tied to a specific class.

A. Continuous Orbit-Level Training with ISL Relay

Under standard FL, the least-visited orbit has almost no influence on the trained model because it participates in only a small fraction of rounds. FedOrbit trains every orbit in every round, using two procedures depending on visibility.

Visible orbits. For $l \in \mathcal{V}(t)$, the number of local epochs grows with the length of the preceding non-visible interval:

$$E_l(t) = \min\bigl(E_{\max},\;\; E_{\mathrm{base}} + s_e\, \tau_l(t)\bigr), \tag{6}$$

where $\tau_l(t)$ is the number of consecutive non-visible rounds prior to t, $s_e \in (0,1)$ scales the catch-up, and $E_{\max}$ caps the load. Each of the R_{intra} intra-orbit rounds uses $E_l(t)$ local epochs and is followed by an intra-orbit aggregation step (Section V-B).

Non-visible orbits (ISL relay). For $l \notin \mathcal{V}(t)$, each satellite in the orbit runs E_{dark} epochs of local SGD on the orbit-level model state, after which an intra-orbit aggregation produces an updated orbit model. This step uses only intra-orbit ISLs and does not consume the GS uplink. E_{dark} is kept small (two epochs) so non-visible refinements do not overwrite the heavier visible updates.

B. Class-Aware Hierarchical Aggregation

When a class is concentrated on a rarely visible orbit, ordinary averaging dilutes the corresponding classifier row with updates from orbits that do not hold the class. FedOrbit weights aggregation by per-class data ownership at both the intra-orbit and inter-orbit tiers.

Class strength and class affinity. For satellite $k \in \mathcal{S}_l$ and class $c \in \{1, \ldots, C\}$,

$$s_{k,c} = \bigl|\{x \in \mathcal{D}_k : y(x) = c\}\bigr|, \tag{7}$$

and the class affinity of orbit l for class c is

$$a_{l,c} = \frac{\sum_{k\in\mathcal{S}_l} s_{k,c}}{\sum_{j=1}^{L}\sum_{k\in\mathcal{S}_j} s_{k,c}}. \tag{8}$$

Both quantities are computed once at initialisation and remain fixed throughout training.

Intra-orbit aggregation. The classifier row for class c is averaged across the orbit's satellites by class strength, and the feature extractor by the magnitudes of the per-satellite updates $\Delta\mathbf{w}_k^{\mathrm{f}} = \mathbf{w}_k^{\mathrm{f}} - \mathbf{w}^{\mathrm{f}}$:

$$\mathbf{w}_l^{\mathrm{c}}[c,:] = \sum_{k\in\mathcal{S}_l} \frac{s_{k,c}}{\sum_{j\in\mathcal{S}_l} s_{j,c}}\, \mathbf{w}_k^{\mathrm{c}}[c,:], \tag{9}$$

$$\mathbf{w}_l^{\mathrm{f}} = \sum_{k\in\mathcal{S}_l} \frac{\bigl|\Delta\mathbf{w}_k^{\mathrm{f}}\bigr|}{\sum_{j\in\mathcal{S}_l}\bigl|\Delta\mathbf{w}_j^{\mathrm{f}}\bigr|}\; \mathbf{w}_k^{\mathrm{f}}, \tag{10}$$

where $|\cdot|$ is the elementwise absolute value, and the ratio in (10) is elementwise.

Inter-orbit aggregation of the classifier. At the GS, the classifier row for class c is averaged across all L orbits with weights given by class affinity times a staleness factor:

$$\mathbf{w}^{\mathrm{c}}[c,:] \leftarrow \frac{\sum_{l=1}^{L} a_{l,c}\, \rho^{\sigma_l(t)}\, \mathbf{w}_l^{\mathrm{c}}[c,:]}{\sum_{l=1}^{L} a_{l,c}\, \rho^{\sigma_l(t)}}, \tag{11}$$

with $\rho \in (0,1]$ and $\sigma_l(t)$ the number of rounds since orbit l last updated its classifier; $\sigma_l(t)$ is small because every orbit trains in every round.

C. Quality-Weighted Feature Aggregation and Return-Rate Dampening

Continuous training produces orbit-level updates with different training budgets: a visible orbit returning from a long non-visible interval performs up to $E_{\max} \cdot R_{\text{intra}}$ epochs, while a non-visible orbit performs only E_{dark}. FedOrbit weights inter-orbit feature aggregation by the amount of training each orbit performed:

$$\bar{\mathbf{w}}^{\text{f}}(t) = \frac{\sum_{l=1}^{L} q_l(t)\, \mathbf{w}_l^{\text{f}}}{\sum_{l=1}^{L} q_l(t)}, \tag{12a}$$

$$q_l(t) = \begin{cases} E_l(t)\, R_{\text{intra}}, & l \in \mathcal{V}(t), \\ E_{\text{dark}}, & l \notin \mathcal{V}(t). \end{cases} \tag{12b}$$

Under the configuration used in our experiments ($E_{\text{base}} = 5$, $E_{\max} = 10$, $R_{\text{intra}} = 2$, $E_{\text{dark}} = 2$), a visible orbit contributes a weight of 10 to 20 and a non-visible orbit a weight of 2.

A returning orbit can train at the global learning rate for up to $E_{\max}$ epochs in each intra-orbit round, producing a feature shift that other orbits' classifier rows have not adapted to. To attenuate this, FedOrbit dampens the local rate of any orbit re-entering visibility in proportion to the length of the non-visible interval that just ended:

$$\eta_l(t) = \frac{\eta_t}{1 + \kappa\, \tau_l(t)/\tau^{\max}}, \tag{13}$$

where $\tau^{\max} = 191$ rounds is the longest non-visible interval and $\kappa \geq 0$ is a hyperparameter. For $\kappa = 0.5$, an orbit returning after the longest interval uses $\eta_l(t) = \eta_t/1.5$, two thirds of the global value; an orbit visible in the previous round ($\tau_l(t) = 0$) is not dampened. The rule is applied only on the first round after re-entry.

D. Adaptive Feature Decomposition

The aggregation rules above produce a single global $[\bar{\mathbf{w}}^{\text{f}}(t), \mathbf{w}^{\text{c}}(t)]$. A single global model is appropriate when class distributions overlap and suboptimal when they are disjoint. FedOrbit adapts between these two cases using a single coefficient computed from the data.

For each orbit l, let $\mathbf{p}_l \in \Delta^{C-1}$ be the normalised class histogram aggregated over its satellites, $p_{l,c} = \sum_k s_{k,c} / \sum_{c',k} s_{k,c'}$. The Jensen-Shannon distance between orbits i and j is

$$\text{JS}(\mathbf{p}_i, \mathbf{p}_j) = \sqrt{\tfrac{1}{2} D_{\text{KL}}(\mathbf{p}_i \| \mathbf{m}) + \tfrac{1}{2} D_{\text{KL}}(\mathbf{p}_j \| \mathbf{m})}, \tag{14}$$

with $\mathbf{m} = \frac{1}{2}(\mathbf{p}_i + \mathbf{p}_j)$ and D_{KL} using $\log_2$, so JS $\in [0, 1]$. The inter-orbit similarity coefficient is

$$\beta = \frac{2}{L(L-1)} \sum_{i<j} \big(1 - \text{JS}(\mathbf{p}_i, \mathbf{p}_j)\big). \tag{15}$$

The coefficient is computed once and fixed.

Algorithm 1 FedOrbit: one communication round.

Require: $\mathbf{V}$; $\{s_{k,c}\}$, $\{a_{l,c}\}$ from (7), (8); β from (15); $\tau^{\max}$

State: $\mathbf{w}^{\text{c}}(t)$, $\bar{\mathbf{w}}^{\text{f}}(t)$, $\{\mathbf{w}_l^{\text{f,p}}(t)\}$, $\{\tau_l, \sigma_l\}$; compute $\mathcal{V}(t)$
for all $l \in \mathcal{V}(t)$ ▷ visible **do**
  Set $E_l(t)$ via (6); $\eta_l(t)$ via (13); init from (17)
  for $r = 1, \ldots, R_{\text{intra}}$ **do**
    Each $k \in \mathcal{S}_l$ runs $E_l(t)$ epochs at $\eta_l(t)$; aggregate via (9), (10)
  end for
  $q_l = E_l(t) R_{\text{intra}}$; reset τ_l, σ_l
end for
for all $l \notin \mathcal{V}(t)$ ▷ non-visible **do**
  Init from (17); each k runs E_{dark} epochs at η_t; aggregate via (9),(10)
  $q_l = E_{\text{dark}}$; τ_l, σ_l += 1
end for
GS: $\bar{\mathbf{w}}^{\text{f}}(t) \leftarrow$ (12); $\mathbf{w}^{\text{c}}(t{+}1) \leftarrow$ (11); $\mathbf{w}_l^{\text{f,p}}(t{+}1) \leftarrow$ (16)
return $\{\widetilde{\mathbf{w}}_l(t+1)\}$ via (17)

Each orbit l maintains a personal feature extractor $\mathbf{w}_l^{\text{f,p}}$, initialised at $\mathbf{w}_l^{\text{f,p}}(0) = \mathbf{w}^{\text{f}}(0)$ and updated by an exponential blend toward the global quality-weighted average:

$$\mathbf{w}_l^{\text{f,p}}(t+1) = (1-\beta)\, \mathbf{w}_l^{\text{f,p}}(t) + \beta\, \bar{\mathbf{w}}^{\text{f}}(t). \tag{16}$$

The same β is used for all orbits and rounds. The orbit's inference model is the pair of its personal feature extractor and the global classifier:

$$\widetilde{\mathbf{w}}_l(t) = \big[\mathbf{w}_l^{\text{f,p}}(t),\ \mathbf{w}^{\text{c}}(t)\big]. \tag{17}$$

A larger β moves the personal extractor toward the global one faster; a smaller β preserves orbit-specific information. Both partitions are handled by the same procedure with the same hyperparameters.

E. Communication and Computation Cost

Each visible orbit uploads one orbit-level model to the GS, as in FedAvg. FedOrbit adds intra-orbit ISL exchanges and local training during non-visible periods. Both operations are performed in parallel across orbits. For a model with P parameters, an FP32 model requires approximately $4P$ bytes; training also requires memory for gradients, momentum, and activations. The sub-million-parameter LeNet models used here therefore require tens of megabytes of working memory. A practical deployment would require an on-board multicore processor supported by an FPGA, GPU, or AI accelerator.

At the GS, aggregation and personal-model storage grow linearly with L, while the one-time similarity

TABLE I
TRAINING HYPERPARAMETERS

Optimiser	SGD, mom. 0.9	E_{base}	5
Weight decay	5×10^{-4}	$E_{\max}$	10
Grad. clip	1.0	E_{dark}	2
Batch	64	s_e	0.7
η_0	0.01	R_{intra}	2
γ	0.998	ρ, κ	0.95, 0.5

calculation in (15) has complexity $\mathcal{O}(L^2C)$. The visibility matrix $\mathbf{V}$ supports time-varying connectivity, but larger and dynamically reconfigured constellations have not been evaluated. These settings and hardware-based validation are left for future work.

VI. EXPERIMENTAL SETUP

Constellation and visibility. All experiments use the Walker Delta configuration of Section III-A. The visibility matrix $\mathbf{V}$ is precomputed via (1) and used by every method. Each simulation uses $T = 200$ communication rounds for EuroSAT and So2Sat LCZ42 and $T = 400$ for RESISC45.

Datasets. EuroSAT [25]: Sentinel-2 land use, 10 classes, 64×64 RGB, 27,000 samples. So2Sat LCZ42 [26]: Sentinel-2 Local Climate Zone, 10 urban classes, 32×32, 10 bands, approximately 352,000 training and 24,000 test samples. NWPU-RESISC45 [27]: aerial scenes, 45 classes, 32×32 RGB, 700 images per class, 80/20 split.

Non-IID partitioning. Both partitions in Section III-C are evaluated. The test samples are distributed among orbits using the corresponding partitioning rule.

Model and training. The on-board model is LeNet (two 5×5 convolutional layers of width 6 and 16 with 2×2 max-pooling, followed by fully connected layers `fc1`, `fc2`, `fc3` of widths 120, 84, C). Input channels are 3 for EuroSAT and RESISC45 and 10 for So2Sat. The feature/classifier split is between `fc1` and `fc2`, and the class-affinity weighting (11) is applied to the rows of `fc3`. Training hyperparameters (Table I) are identical across methods within each dataset.

Baselines. FedAvg [5] averages visible-orbit models (4); FedProx [10] adds a proximal term ($\mu = 0.01$); APFL [14] interpolates the FedAvg global and a local orbit model with an adaptively learned mixing coefficient; Ditto [13] trains a per-orbit personal model with an ℓ_2 proximal term ($\lambda_{\text{Ditto}} = 0.1$) toward a FedAvg global trained in parallel. The optimiser, batch size, initial learning rate, and scheduler are shared across methods.

Evaluation. For each orbit l, Acc_l is the accuracy of $\widetilde{\mathbf{w}}_l$ on orbit l's test partition (the global model for FedAvg and FedProx, the personalised model for APFL and Ditto, and (17) for FedOrbit). The primary metric is the per-orbit mean $\frac{1}{L}\sum_l \mathrm{Acc}_l$, averaged over the last ten evaluated rounds; the secondary metric is the per-orbit spread $\max_l \mathrm{Acc}_l - \min_l \mathrm{Acc}_l$. All methods were implemented in PyTorch using the same partitions and visibility matrices in each repetition.

The hyperparameters in Table I are fixed across the three datasets and both partitions. Adaptation to heterogeneity is provided by β in (15), not by dataset-specific tuning. A broader sensitivity analysis of κ, ρ, and s_e is left for future work.

TABLE II
PERSONALISED ACCURACY (TOP) AND PER-ORBIT SPREAD (BOTTOM)

	EuroSAT		**So2Sat**		**RESISC45**	
Method	Dir.	Path.	Dir.	Path.	Dir.	Path.
Accuracy						
FedAvg	68.7	20.0	38.9	16.9	37.3	14.8
FedProx	68.5	20.3	39.1	16.9	38.1	15.4
APFL	60.9	74.7	39.9	40.7	32.3	54.4
Ditto	58.5	**95.8**	35.7	84.0	36.3	77.7
FedOrbit	**76.4**	94.9	**50.9**	**86.9**	**54.2**	**86.3**
Spread						
FedAvg	28.4	78.1	26.1	68.2	10.0	53.3
FedProx	31.0	78.2	27.9	68.4	9.9	55.4
APFL	40.2	53.9	31.6	70.2	12.1	38.4
Ditto	49.0	**9.7**	32.6	17.6	11.6	17.6
FedOrbit	**15.8**	12.5	**24.7**	**15.8**	**8.7**	**8.9**

VII. RESULTS AND ANALYSIS

Table II reports the per-orbit personalised accuracy and the per-orbit accuracy spread on all three datasets under both partitions. FedOrbit achieves the highest accuracy in five of six settings and is within 0.9 percentage points of the best result in the sixth.

Pathological partition. With disjoint inter-orbit classes, FedAvg and FedProx achieve only 14.8% to 20.0% because the global classifier serves only the classes on frequently visible orbits. APFL improves through interpolation but is regularised toward the same biased anchor. Ditto is the strongest baseline; FedOrbit gains 8.6 percentage points over Ditto on RESISC45 and 2.9 on So2Sat, while trailing Ditto by 0.9 percentage points on EuroSAT.

Dirichlet partition. A different pattern is observed. Ditto is 37 to 48 percentage points lower than under pathological partitioning, and APFL also degrades because strong personalisation is less beneficial when orbit distributions overlap. FedOrbit gains 7.7, 11.0, and 16.1 percentage points over the strongest baseline on EuroSAT, So2Sat, and RESISC45. The adaptive feature decomposition provides a consistent explanation: β (15) is moderate under Dirichlet partitioning (≈ 0.44) and small under pathological partitioning (≈ 0.11). The personal feature extractor therefore tracks the global average more quickly under Dirichlet partitioning and retains orbit-specific information for longer under pathological partitioning.

Per-orbit fairness. The spread block of Table II operationalises the fairness term in (5). FedOrbit has the smallest spread in five of six settings and is within 3 percentage points of the smallest in the sixth (EuroSAT pathological). Under pathological partitioning, the global baselines leave spreads of 53 to 78 percentage points, confirming poor accuracy for low-visibility orbits.

Component interpretation. As the mechanisms operate jointly, Table II does not isolate their individual effects. Fixed hyperparameters avoid dataset-specific tuning, while the change in β is consistent with stronger sharing under Dirichlet partitioning and greater orbit-specific retention under pathological partitioning. A controlled ablation is left for future work.

VIII. Conclusion

FedOrbit addresses the coupling between orbit-level data heterogeneity and visibility-dependent participation in LEO constellations. It combines continuous orbit-level training, class-aware hierarchical aggregation, quality-weighted feature aggregation with return-rate dampening, and adaptive feature decomposition. Across three remote-sensing datasets, FedOrbit achieves the highest accuracy in five of six settings and is within 0.9 percentage points of the best result in the sixth. It also produces the smallest per-orbit accuracy spread in five of six settings. Future work will evaluate individual component contributions, hardware deployment, and larger dynamic constellations.

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